Year 10 Foundation

REVISION PACK

Number 3

Name: _____

Class: _____

Number 3 · Laws of indices (Lesson 6.1)

Worked example: writing as a single power

Write as a single power of the number.

(a) $6^3 \times 6^4$ (b) $9^8 \div 9^2$ (c) $(2^3)^4$

(a)	**Step 1**	Multiply: add the powers	$3 + 4 = 7$
	Step 2	Answer	6^7
(b)	**Step 1**	Divide: subtract the powers	$8 - 2 = 6$
	Step 2	Answer	9^6
(c)	**Step 1**	Power of a power: multiply	$3 \times 4 = 12$
	Step 2	Answer	2^{12}

Practice: writing as a single power

1.1 Write as a single power of the number.

(a) $4^5 \times 4^2$ (b) $8^{10} \div 8^3$ (c) $(5^2)^3$

1.2 Write as a single power of the number.

(a) $6^2 \times 6^7$ (b) $9^8 \div 9^5$ (c) $(7^4)^2$

1.3 Write as a single power of the number.

(a) $3^4 \times 3^6$ (b) $2^9 \div 2^2$ (c) $(2^5)^3$

1.4 Write as a single power of the number.

(a) $10^3 \times 10^5$ (b) $5^{12} \div 5^4$ (c) $(4^3)^4$

1.5 Write as a single power of the number.

(a) $7^2 \times 7^8$ (b) $3^{11} \div 3^6$ (c) $(6^2)^6$

1.6 Write as a single power of the number.

(a) $2^6 \times 2^5$ (b) $6^9 \div 6^3$ (c) $(9^3)^3$

Number 3 · Laws of indices (Lesson 6.1)

Worked example: negative and zero powers

Work out the value of

(a) 3^{-2} (b) 8^0

(a)	Step 1	Minus power: 1 over	$3^{-2} = \frac{1}{3^2}$
	Step 2	Answer	$\frac{1}{9}$
(b)	Step 1	Power 0 is always 1	$8^0 = 1$
	Step 2	Answer	1

Practice: negative and zero powers

1.7 Work out the value of

(a) 4^{-2} (b) 7^0

1.8 Work out the value of

(a) 2^{-4} (b) 25^0

1.9 Work out the value of

(a) 10^{-2} (b) 9^0

1.10 Work out the value of

(a) 3^{-3} (b) 15^0

1.11 Work out the value of

(a) 5^{-2} (b) 100^0

1.12 Work out the value of

(a) 6^{-1} (b) 48^0

Number 3 · Using index laws (Lesson 6.1)

Worked example: working out with powers

Work out

(a) $5^2 \times 5^4$. Give your answer as a power of 5. (b) the value of $2^5 \div 2$ (c) the value of 7^0

(a)	**Step 1**	Add the powers	$2 + 4 = 6$
	Step 2	Answer	5^6
(b)	**Step 1**	Subtract the powers	$2^{5-1} = 2^4$
	Step 2	Work out 4^2	$2^4 = 16$
	Step 3	Answer	16
(c)	**Step 1**	Power 0 is always 1	$7^0 = 1$
	Step 2	Answer	1

Practice: working out with powers

2.1 Work out

(a) $2^4 \times 2^3$. Give your answer as a power of 2. (b) the value of $5^3 \div 5$ (c) the value of 9^0

2.2 Work out

(a) $4^2 \times 4^5$. Give your answer as a power of 4. (b) the value of $3^4 \div 3$ (c) the value of 12^0

2.3 Work out

(a) $6^3 \times 6^2$. Give your answer as a power of 6. (b) the value of $2^5 \div 2$ (c) the value of 50^0

2.4 Work out

(a) $7^5 \times 7^2$. Give your answer as a power of 7. (b) the value of $10^4 \div 10$ (c) the value of 3^0

2.5 Work out

(a) $5^2 \times 5^6$. Give your answer as a power of 5. (b) the value of $6^3 \div 6$ (c) the value of 8^0

2.6 Work out

(a) $3^5 \times 3^4$. Give your answer as a power of 3. (b) the value of $2^6 \div 2$ (c) the value of 11^0

Number 3 · Finding the power (Lesson 6.1)

Worked example: finding the power n

$3^n = \frac{1}{27}$ Find the value of n.

Step 1	Write 27 as a power of 3	$27 = 3^3$
Step 2	One over a power is a negative power	$\frac{1}{27} = 3^{-3}$
Step 3	Answer	$n = -3$

Practice: finding the power n

3.1 $2^n = 32$ Find the value of n.

3.2 $5^n = \frac{1}{25}$ Find the value of n.

3.3 $2^n = \frac{1}{8}$ Find the value of n.

3.4 $10^n = 0.001$ Find the value of n.

3.5 $4^n = 1$ Find the value of n.

3.6 $3^n = \frac{1}{81}$ Find the value of n.